

Quadratic Simultaneous Equations

AQA Higher · Grade 8–9

Solve each pair. Show your working and answers in your workbook.

Method

- 1 Make y the subject of the straight line
- 2 Substitute into the quadratic
- 3 Rearrange so it = 0
- 4 Solve (factorise or formula)
- 5 Find each pair (x, y) and check

Common mistakes

- There are usually two answers, not one — find both x values, e.g. $x = 3$ and $x = -1$.
- Get everything on one side so it = 0 first, e.g. $x^2 = 2x + 3$ becomes $x^2 - 2x - 3 = 0$.
- Watch minus signs: $(x - 1)^2 = x^2 - 2x + 1$, not $x^2 - 1$.
- Each x has its own y — put x back in to find y .

Worked example

Solve: $y = x^2 - 2x + 3$ and $y = x + 3$

Step 1 — Make y the subject of the line: $y = x + 3$

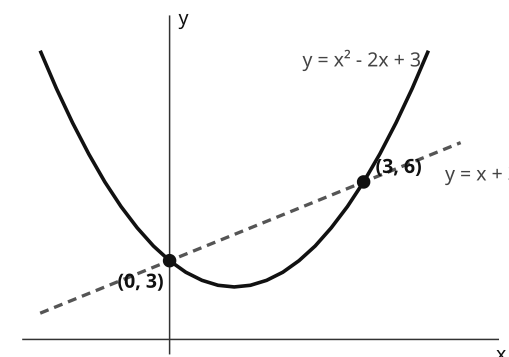
Step 2 — Substitute into the quadratic: $x^2 - 2x + 3 = x + 3$

Step 3 — Rearrange so it = 0: $x^2 - 3x = 0$

Step 4 — Solve (factorise): $x(x - 3) = 0 \rightarrow x = 0$ or $x = 3$

Step 5 — Find each pair (x, y) and check: $(0, 3)$ and $(3, 6)$

Visual aid — two solutions



The straight line meets the curve at the two answer points.

Questions

Solve each pair — show working in your workbook

Q1 of 8

[3] ● ○ ○ ○ ○

$$y = x^2 \text{ and } y = 2x + 3$$

Q2 of 8

[3] ● ○ ○ ○ ○

$$y = x^2 + 1 \text{ and } y = x + 3$$

Q3 of 8

[3] ● ● ○ ○ ○

$$y = x^2 - 2x \text{ and } y = x + 4$$

Q4 of 8

[3] ● ● ○ ○ ○

$$y = x^2 - 5x + 6 \text{ and } y = x - 2$$

Q5 of 8

[4] ● ● ● ○ ○

$$x^2 + y^2 = 25 \text{ and } y = x - 1$$

Q6 of 8

[4] ● ● ● ○ ○

$$x^2 + y^2 = 10 \text{ and } x + y = 4$$

Q7 of 8

[5] ● ● ● ● ○

$$x^2 + y^2 = 9 \text{ and } y = x + 1 \text{ (give answers in surd form)}$$

Q8 of 8

[5] ● ● ● ● ●

A rectangle has area 12 cm^2 and perimeter 14 cm . Form and solve a pair of equations to find its length (x) and width (y).